Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Technical solutions engineer with **8+ years building and debugging inside a property management platform** - I know PMS integration layers from the source code. Built 20+ production modules at Entrata (100K+ MAU), optimized 200+ SQL queries (30% performance gain), and built internal debugging tools that reduced investigation time 40%. Also built **Leasel** - an 8-agent AI platform for regulated compliance operations - so I understand how AI assistants fail and how to configure them correctly. Taught AI fundamentals and Python to 600+ students at George Mason University. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

PMS Integration & Troubleshooting | SQL Investigation & Root-Cause Analysis | Customer-Facing Technical Advisory | AI Agent Configuration & Debugging | Log Analysis & Issue Documentation | Cross-Functional Stakeholder Communication | Agentic AI & LangGraph | Production AI Systems

WORK EXPERIENCE

Entrata - PropTech / Real Estate SaaS (100K+ MAU)

Sep 2018 - Aug 2024

Senior Software Engineer

- **Built internal debugging and performance tools** reducing debugging time 40% - root-cause analysis across complex multi-schema PostgreSQL, MySQL, and Oracle production environments; maintained bug count below 10 for 6+ consecutive months.
- **Optimized 200+ SQL queries** (30% performance gain) through execution plan tuning, indexing, database partitioning, and caching strategies - primary investigator for performance incidents across all client environments.
- **Shipped 20+ full-stack production modules** (PHP, Python, React, PostgreSQL, RabbitMQ, AWS) across 120+ Agile sprints for 100K+ monthly active users - 99.5% uptime; collaborated cross-functionally with compliance, operations, and product stakeholders to translate complex system behavior into actionable reports.
- **Built regulated data pipelines** on 1.6M financial records - HUD, LIHTC, HCV compliance - with output validation gates, audit trails, and data quality controls; 100% PHI compliance on AWS S3 migration; zero security incidents.
- Led microservices redesign (Docker) - 50% scalability improvement; mentored 5+ engineers across 300+ peer code reviews; eliminated 80% of manual workload via Python/Bash automation.

George Mason University - College of Engineering

Aug 2025 - May 2026

Graduate Teaching Assistant

- Translated complex AI architectures and database concepts into hands-on exercises for **600+ students** - Python programming, AI fundamentals, SQL and database design; one-on-one debugging support and rubric-aligned grading across two subject areas.

Magizhs Tech Solution

May 2025 - Aug 2025

AI Engineer (Intern)

- Architected and deployed a LangGraph agentic system with RAG retrieval, confidence-scored outputs, and validation guardrails across three commercial verticals - self-directed from requirements through production.

George Mason University - Research

Aug 2024 - May 2025

Research Associate - AI & Data Analytics

- Built multi-source prediction models (Python, R) across 190+ countries - R^2=0.977; Isolation Forest anomaly detection surfacing data quality issues across 3 research pipelines; Tableau and ggplot2 dashboards for faculty leadership (40% comprehension improvement).

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer / Trainee

- Built full-stack property management modules (PHP, Python, PostgreSQL); ETL pipelines on 1.6M regulated records; AWS S3 cloud migration.

PROJECTS

Leasel: 8-Agent AI Platform for Regulated Compliance Operations 2026

Built an 8-agent LangGraph system for affordable housing compliance - the same pattern as EliseAI's AI assistant: agents extract, classify, and reason over documents; deterministic logic and human review handle every regulated decision. Configuration errors surface via confidence scoring before reaching users. RAG retrieval (91% Precision@1, 100% citation correctness), LLM-as-judge eval harness, regression gating. Production precision: 93.6% across 1,240 enterprise documents.

Python · LangGraph · FastAPI · RAG · Confidence Scoring · LLM-as-Judge · Eval Harness · SQLite

EDUCATION

M.S. Data Analytics Engineering · George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering · Jawaharlal Nehru Technological University 2017

SKILLS

Languages: Python, SQL, PL/SQL, JavaScript, TypeScript, PHP, R, Bash

Investigation & Troubleshooting: SQL (PostgreSQL, MySQL, Oracle), Root-Cause Analysis, Log Investigation, Execution Plan Tuning, Indexing, Database Partitioning, Data Quality Controls, Audit Trails, Tableau, Pandas, NumPy

AI & Agentic Systems: LangGraph, LangChain, Multi-Agent Orchestration, RAG Pipelines, Confidence Scoring, LLM-as-Judge, Eval Harnesses, FastAPI, ChromaDB, OpenAI API, Anthropic API, spaCy, BERT

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Microservices, CI/CD (GitHub Actions), RabbitMQ, Git, Linux, Agile/Scrum